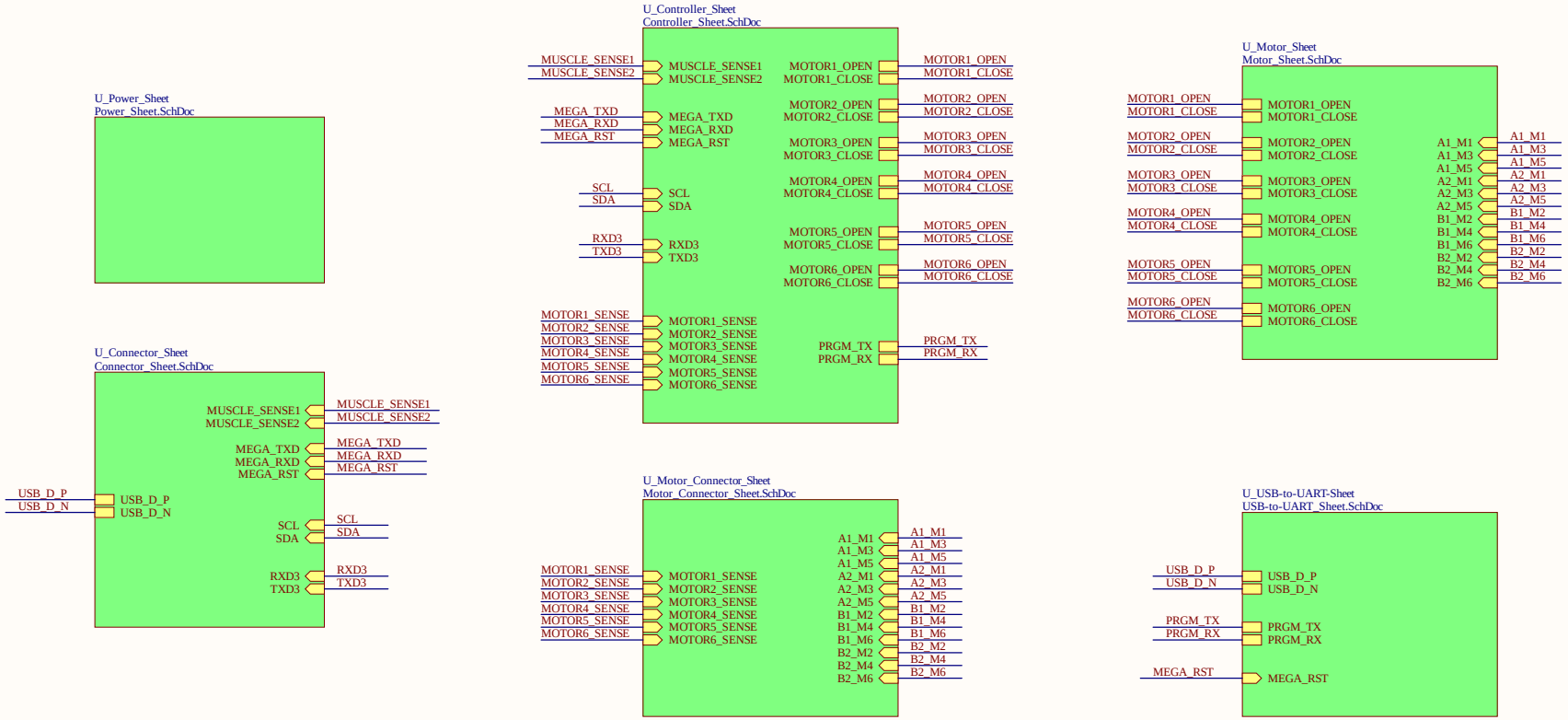
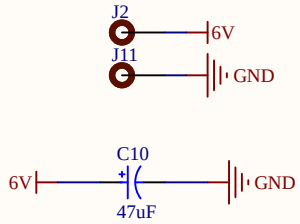
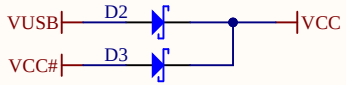

U_Power_Sheet
Power_Sheet.SchDoc
U_Connector_Sheet
Connector_Sheet.SchDoc
MUSCLE_SENSE1
MUSCLE_SENSE2
MEGA_TXD
MEGA_RXD
MEGA_RST
SCL
SDA
RXD3
TXD3
USB_D_P
USB_D_N
U_Controller_Sheet
Controller_Sheet.SchDoc
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
PRGM_TX
PRGM_RX
U_Motor_Sheet
Motor_Sheet.SchDoc
A1_M1
A1_M3
A1_M5
A2_M1
A2_M3
A2_M5
B1_M2
B1_M4
B1_M6
B2_M2
B2_M4
B2_M6
U_Motor_Connector_Sheet
Motor_Connector_Sheet.SchDoc
U_USB-to-UART-Sheet
USB-to-UART_Sheet.SchDoc
MEGA_RST

USB_D_N
USB_D_P

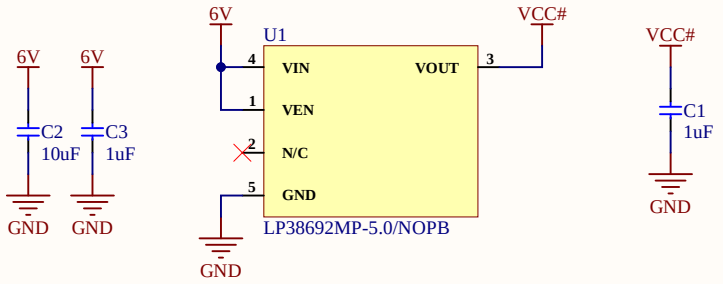
POWER INPUTS



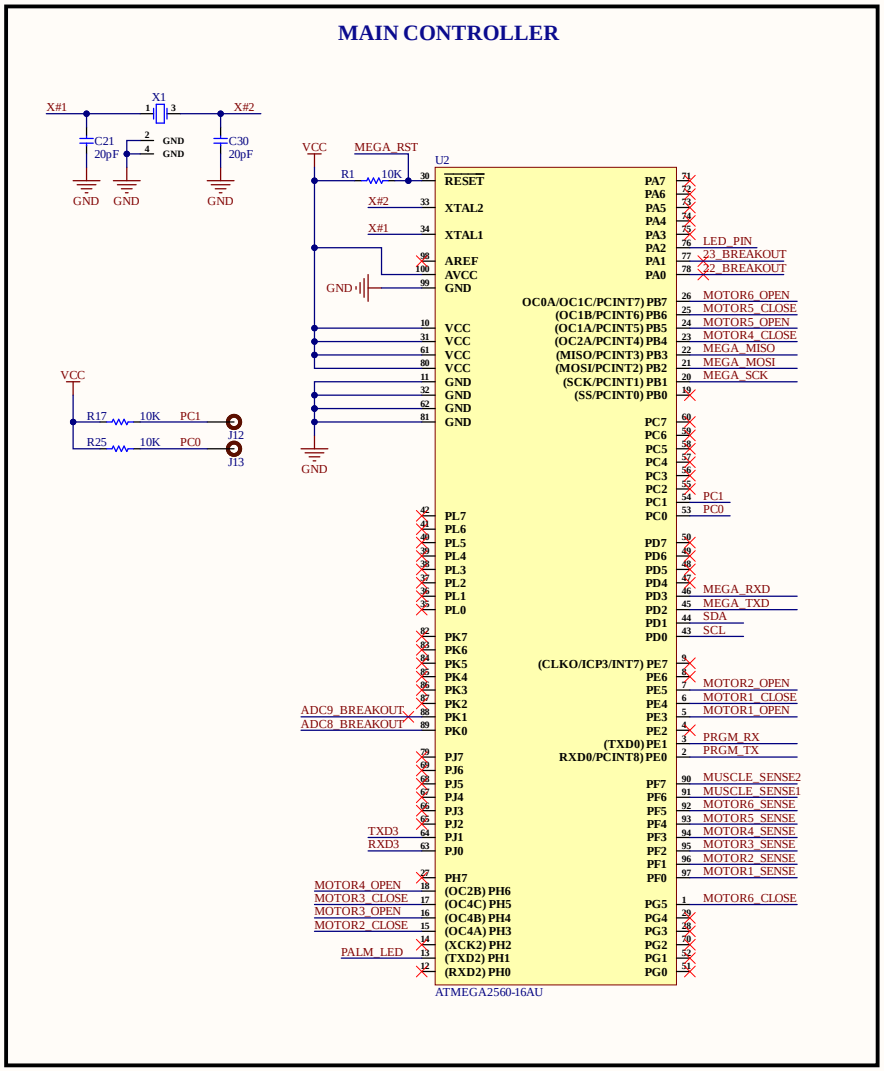
DIODE ORING



6V TO 5V CONVETER



MAIN CONTROLLER



SDA
SCL

TXD3
RXD3

MEGA_TXD
MEGA_RXD
MEGA_RST

PRGM_RX
PRGM_TX

MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE

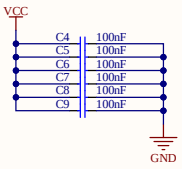
MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE

MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE

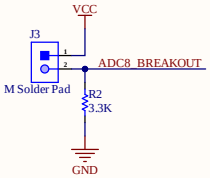
MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE

MUSCLE_SENSE1
MUSCLE_SENSE2

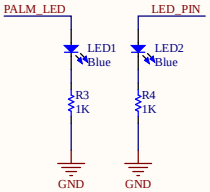
DECOUPLING CAPACITORS



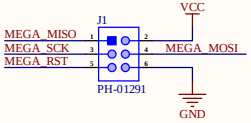
SENSOR CONNECTION



LED'S



SPI



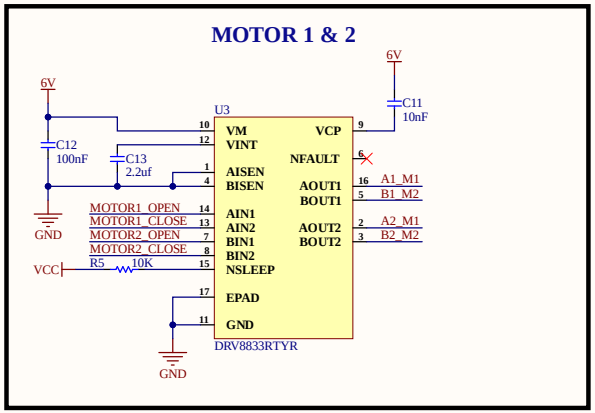
B2_M6
B1_M6
A2_M5
A1_M5
B2_M4
B1_M4
A2_M3
A1_M3
B2_M2
B1_M2
A2_M1
A1_M1

MOTOR1_OPEN
MOTOR1_CLOSE
MOTOR2_OPEN
MOTOR2_CLOSE

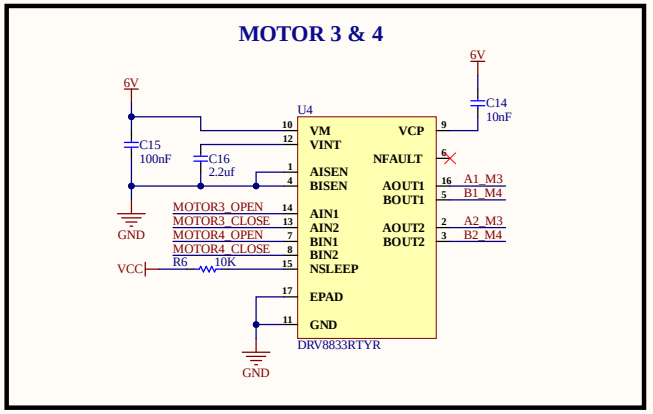
MOTOR3_OPEN
MOTOR3_CLOSE
MOTOR4_OPEN
MOTOR4_CLOSE

MOTOR5_OPEN
MOTOR5_CLOSE
MOTOR6_OPEN
MOTOR6_CLOSE

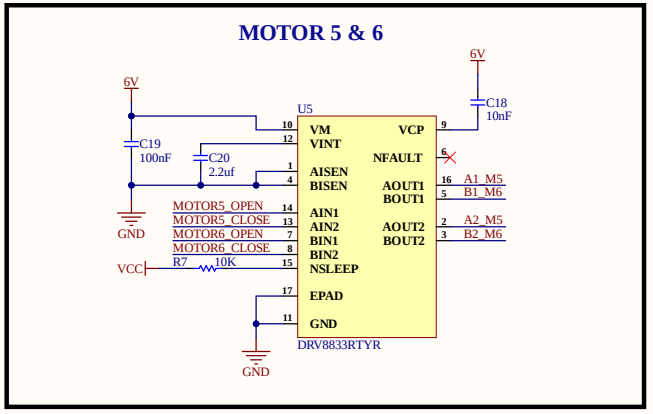
MOTOR 1 & 2

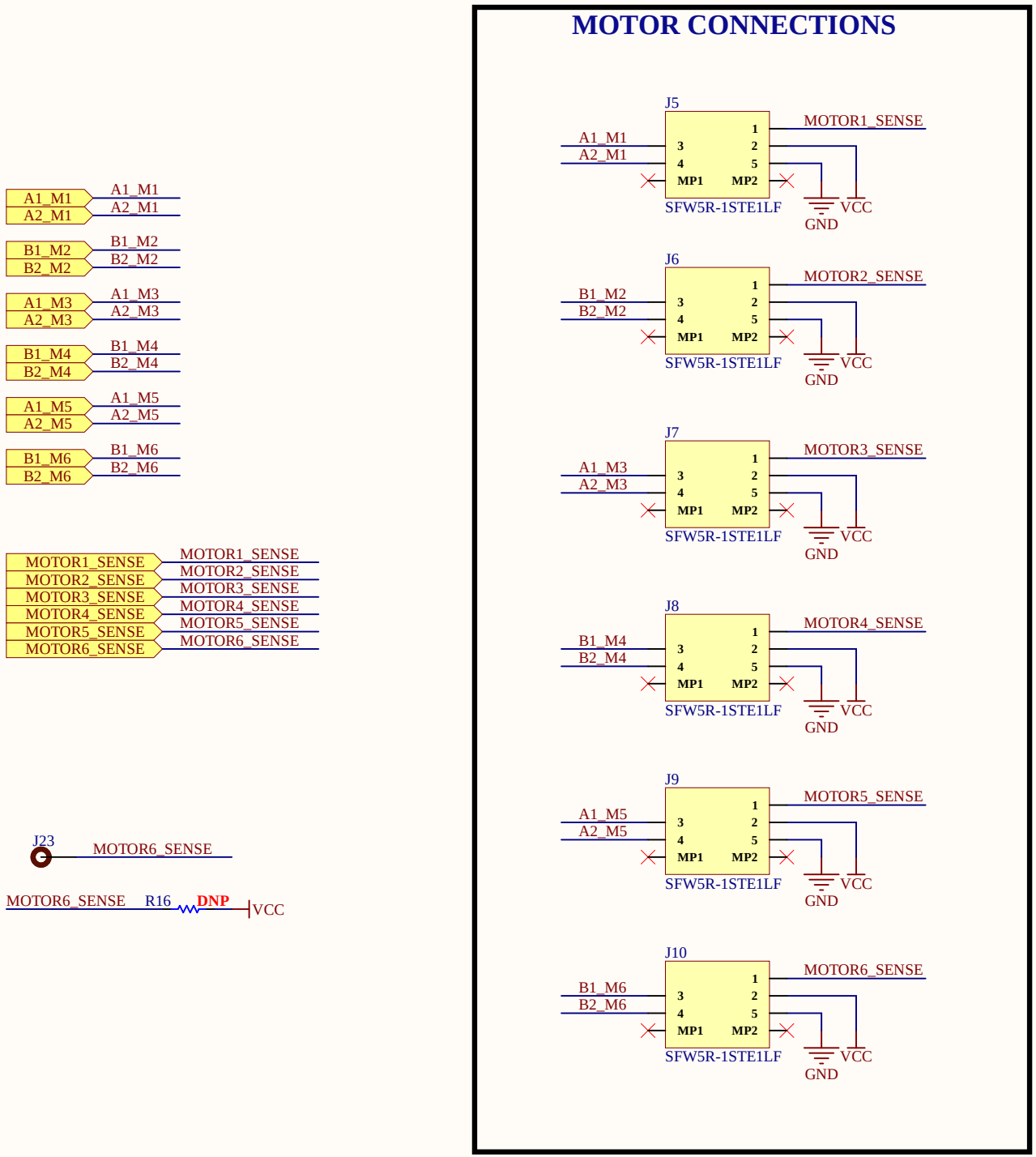


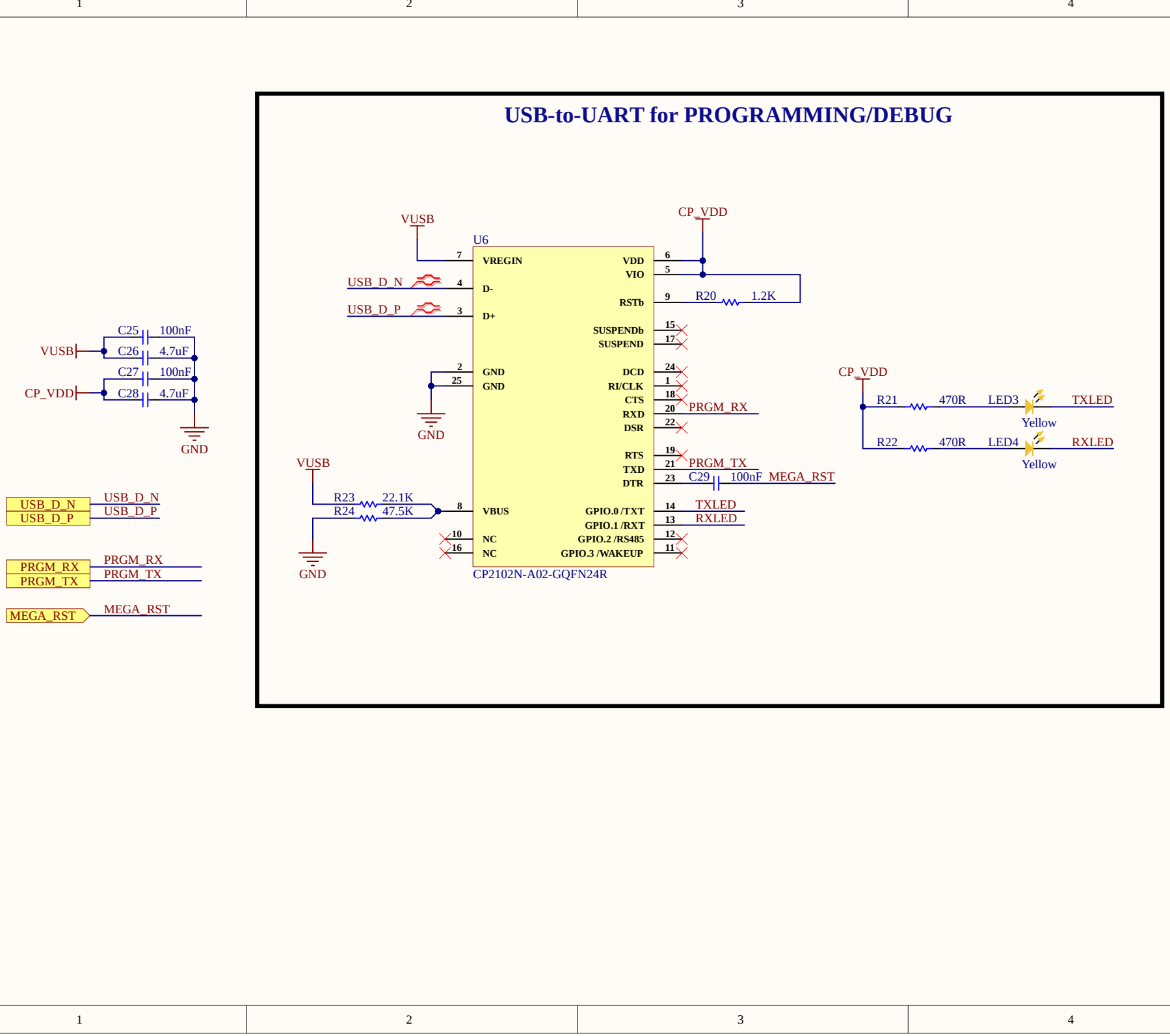
MOTOR 3 & 4



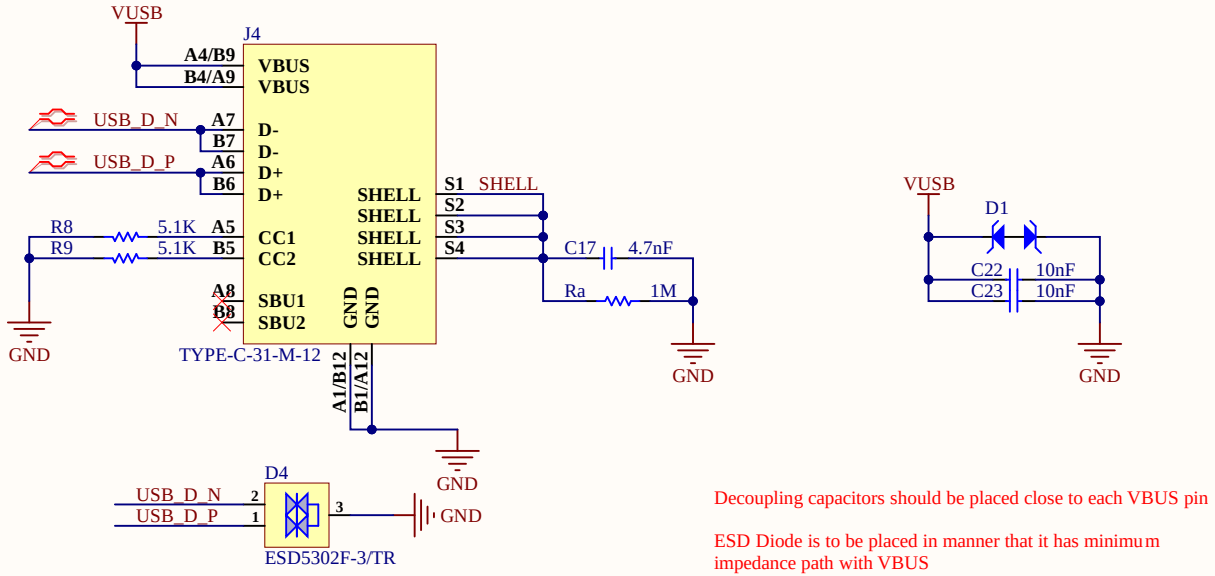
MOTOR 5 & 6




MOTOR CONNECTIONS
J5
A1_M1
A2_M1
MOTOR1_SENSE
SFW5R-1STE1LF
VCC
GND
J6
B1_M2
B2_M2
MOTOR2_SENSE
J7
A1_M3
A2_M3
MOTOR3_SENSE
J8
B1_M4
B2_M4
MOTOR4_SENSE
J9
A1_M5
A2_M5
MOTOR5_SENSE
J10
B1_M6
B2_M6
MOTOR6_SENSE
A1_M1
A2_M1
B1_M2
B2_M2
A1_M3
A2_M3
B1_M4
B2_M4
A1_M5
A2_M5
B1_M6
B2_M6
MOTOR1_SENSE
MOTOR2_SENSE
MOTOR3_SENSE
MOTOR4_SENSE
MOTOR5_SENSE
MOTOR6_SENSE
J23
MOTOR6_SENSE
R16
DNP
VCC


USB-to-UART for PROGRAMMING/DEBUG
VUSB
CP_VDD
U6
VREGIN
VDD
VIO
D-
RSTb
D+
SUSPENDb
SUSPEND
GND
GND
DCD
RI/CLK
CTS
RXD
DSR
RTS
TXD
DTR
VBUS
GPIO.0 /TXT
GPIO.1 /RXT
GPIO.2 /RS485
GPIO.3 /WAKEUP
NC
NC
CP2102N-A02-GQFN24R
USB_D_N
USB_D_P
R20
1.2K
PRGM_RX
PRGM_TX
C29
100nF
MEGA_RST
TXLED
RXLED
R23
22.1K
R24
47.5K
C25
100nF
C26
4.7uF
C27
100nF
C28
4.7uF
R21
470R
LED3
Yellow
R22
470R
LED4
Yellow
MEGA_RST

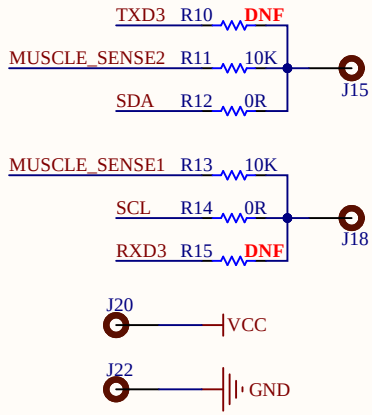
USB-C CONNECTOR

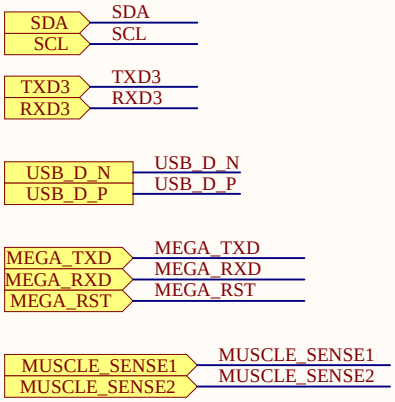


Decoupling capacitors should be placed close to each VBUS pin

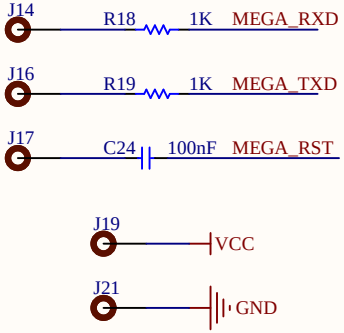
ESD Diode is to be placed in manner that it has minimum impedance path with VBUS

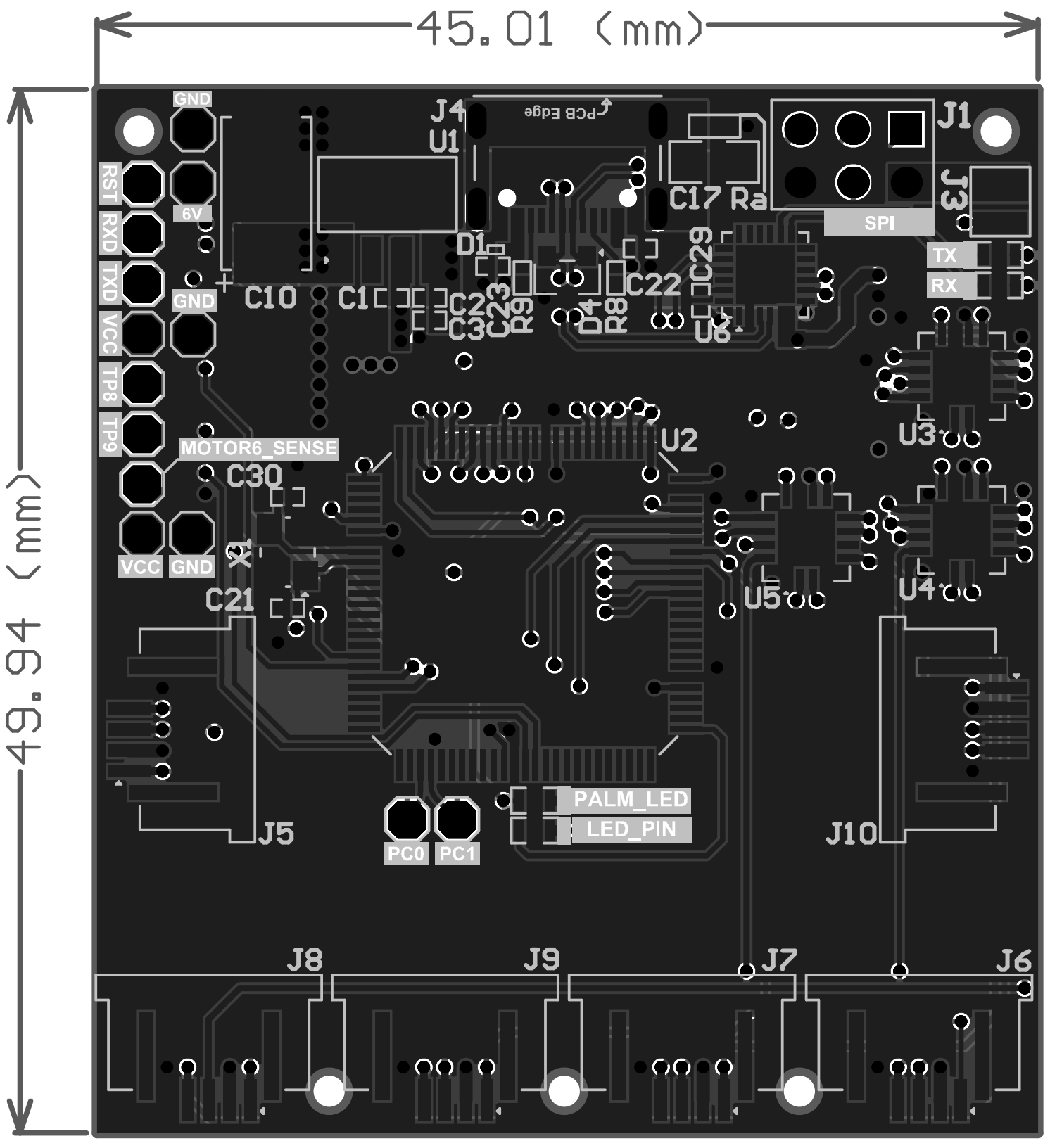
AUDIO CONNECTION





USB CONNECTION




45.01 (mm)
49.94 (mm)
GND
6V
GND
RST
RXD
TXD
VCC
TP8
TP9
VCC
GND
MOTOR6_SENSE
J4
U1
PCB Edge
J1
SPI
J3
TX
RX
C17
Ra
D1
C10
C1
C2
C3
C23
R9
D4
R8
C22
C29
U6
U2
U3
U4
U5
C30
X1
C21
J5
PC0
PC1
PALM_LED
LED_PIN
J10
J8
J9
J7
J6